Artificial intelligence for image-based breast cancer risk prediction using attention

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ABSTRACT

Accurate prediction of individual breast cancer risk paves the way for personalised prevention and early detection. Whilst well-established clinical risk factors can be used to stratify the population into risk groups, the addition of genetic information and breast density has been shown to improve prediction. Machine learning enabled automatic risk prediction provides key advantages over existing methods such as the ability to extract more complex information from mammograms. However, this is a challenging area of research, partly due to the lack of data within the field, therefore there is scope for novel approaches. Our method uses Multiple Instance Learning in tandem with attention in order to make accurate, short-term risk predictions from full-sized mammograms taken prior to the detection of cancer. This approach ensures small features like calcifications are not lost in a downsizing process and the whole mammogram is analysed effectively. An attention pooling mechanism is designed to highlight patches of increased importance and improve performance. Additionally, this increases the interpretability of our model as important patches can be shown in a saliency map. We also use transfer learning in order to utilise a rich source of screen-detected cancers and evaluate whether a model trained to detect cancers in mammograms allows us also to predict risk in priors. Our model achieves an AUC of 0.635 (0.600,0.669) in cancer-free screening mammograms of women who went on to a screen-detected or interval cancer between 5 and 55 months and an AUC of 0.804 (0.777,0.830) in screen-detected cancers.

Keywords: Mammography, Multiple instance learning, risk prediction, deep learning

1. INTRODUCTION

In 2020, female breast cancer became the most commonly diagnosed cancer worldwide,[1] overtaking lung cancer. Various reasons can be attributed to the increased incidence, with lifestyle, environmental, and wider implementation of breast screening all playing a role. Meanwhile continued improvements in breast cancer treatment and early detection have resulted in mortality decreasing.

Screening programmes are designed to screen populations most at risk from cancer and detect malignancies early. The UK National Screening Committee recommends triennial mammography screening starting at 50 years of age until 70,[2] however, guidelines differ between countries.[3] Whilst they largely achieve their goals, screening programmes also have drawbacks including overdiagnosis and differential uptake amongst different groups of women.[4] These issues can all be summarised by a lack of personalised screening and one of the first steps towards personalised screening is the development of better, more custom risk prediction tools.

At the moment clinicians use risk models based on well-known clinical risk factors. Models like the Tyrer-Cuzick (TC)[5,6] work well on large populations but leave room for improvement when it comes to individual risk or short-term risk. This is demonstrated by their high calibration but a moderate discrimination accuracy.[7,8] Furthermore, recent research[9] has started placing increased importance on mammographic appearance for risk assessment, which can be hard to integrate into numeric models like the TC. Mammographic presence in risk

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models can by extension include unrestricted breast disease markers which are normally underrepresented in classic risk models. These features may include pattern and location of dense tissue,[10] the presence of micro-calcifications[11] or other more subtle appearances associated with breast cancer risk independent to volumetric breast density.

Thus our key motivation is to build an accurate, image-based breast cancer risk model, with a specific focus on short-term risk. Deep learning approaches is an area of particular interest since they have proven successful in various medical fields and are excellent at processing image-based information. Through focusing on short-term risk, our models may be more clinically useful, since they could work alongside screening programmes. By using cancer free prior screening mammograms of women with subsequent screen-detected or interval cancers as the basis for learning, the prospect is that our models could detect patients who would go on to develop breast cancer in the next 5 years, hence could have an adjusted screening protocol. This has the promise of further reducing the rates of interval cancer and improving early diagnosis.

However, tackling this challenge has proven difficult. Deep learning models require an adequate amount of labelled data in order to achieve good performance and there is a limited amount of data available to develop image-based risk prediction models. Additionally, deep learning models typically require a significant amount of computational power and would normally downsize images in order to reduce their dimensionality. However, downsizing can lead to loss of fine information such as calcifications which play an important role in risk prediction.[11] Another issue is the lack of explainability when proposing deep learning solutions leading to "black box" models. This problem is only exacerbated for cancer risk prediction since clinicians might be interested in the underlying reason as to why a patient might be classified as high-risk. We want to ensure our models can provide an insight into how risk prediction was made. The nature of transparency is also important given our target audience is clinicians, who might not have formal machine learning training.

1.1 Objectives

In order to address the challenges we have highlighted we are proposing a transfer-learning enabled Multiple Instance Learning (MIL) model with attention-based aggregation. By using multiple instance learning we can convert our images into a group of smaller instances, typically referred to as a *bag*, reducing the computational burden without strict downsizing. Furthermore, saliency maps can be developed based on the instances which can provide a pathway towards an explainable model. However, the essence of our approach lies in transfer learning. Models trained to detect breast cancers in cancer positive screening images are more easily developed as there is more such data available and the scope of the problem is more manageable.

It follows that our primary objective is to answer the question of whether MIL based models trained to detect screen-detected cancers in mammograms can allow us to predict risk in women whose screening mammograms were read as normal, but had cancers detected within the next 5 years.

Our **aims** can be summarised as follows:

- develop a cancer detection model that is able to predict risk in priors
- achieve computational feasibility without downsizing or otherwise rescaling the mammograms
- provide interpretations of the model decision making rationale and investigate the logic behind it

1.2 Literature

Perhaps one of the most famous approaches to breast cancer classification was made by Google's McKinney et al.[12] Their approach used an ensemble of 3 separate deep CNNs and employed a large dataset of 28,953 women from the US and UK. The AI system was able to outperform readings by a single human radiologist whilst having access to less information (clinical data for radiologist vs age only for the AI system) with a 1.2% better specificity and 2.7% better sensitivity. However, the UK uses the double-reader system which the AI was not able to outperform. The advantages of Google's AI include a much larger mammographic dataset than most other approaches and extreme computational power.

Several studies have already recognised the potential of MIL approaches for mammography. Shen et al.[13] used a mixed global and local network to detect malignant lesions. The method used a low capacity global network to extract candidate regions which were subsequently analysed by a higher capacity network and later aggregated using an attention mechanism. Other methods[14,15]have also sought to use MIL based approaches however they have only been utilised for breast cancer detection making them more appropriate for that field rather than risk prediction.

Deep learning based automatic breast cancer risk prediction methods have also been previously proposed. Dembrower et al.[9] investigated whether deep learning based approaches could supplant breast density as a stronger risk prediction factor for future breast cancer. The convolutional neural network (CNN) based approach showed a modest improvement, with an Area Under the receiver operating characteristic Curve (AUC) of 0.65 without age adjustment, however, truly assessing the comparison between density and mammography for risk prediction is difficult because mammography might have early signs of cancer. Similarly, Yala et al.[16] explored deep learning as an avenue for risk prediction. A hybrid method using classic breast cancer risk factors with deep learning extracted features was able to outperform the TC model with the task of identification of a high-risk population. Their methods achieved an AUC of 0.68 using mammography images only, and a further AUC of 0.70 with a hybrid model incorporating clinical data. Another notable approach[17] used deep learning methodology to find percentage breast density which was then used to calculate overall cancer risk.

Whilst these methods were able to achieve impressive results there are several pitfalls that we hope to improve with our approach to the automatic risk prediction problem. Firstly we hope to use the full size of the mammogram without significant computational burden since downsizing the image might erase certain detail within the breast. The method by Yala et al.[16] already uses the full size mammogram but their dataset has a much smaller native dimension when comparing to our data. More importantly, none of these methods provide interpretable answers as to how the prediction was made, a growing issue in the medical imaging field which we hope to alleviate.

2. DATA

2.1 Dataset

Our data is a subset of the Predicting Risk of Breast Cancer at Screening (PROCAS)[18] study in Greater Manchester. It consists of entry mammograms of around 45000 women. Each woman should have the 4 standard screening views, craniocaudal (CC) and mediolateral oblique (MLO) for each breast. All mammograms have one of two dimensions, 3062×2394 or 2294×1914 with equal pixel sizes for both types. The 2 relevant datasets for model development are the **PROCAS-priors** and the **PROCAS-sdc** (screen-detected cancer).[18] PROCAS-priors consists of 320 women confirmed cancer free at screening who went on to develop a screen-detected or interval cancer between 5 and 55 months. PROCAS-sdc has the same basic format as PROCAS-priors but includes women with screen-detected cancer. This dataset features 370 positives. Every positive case in both datasets is matched against 3 controls based on age, BMI category, menopausal status and parity. Exclusion criteria included women with previous breast cancer diagnosis and women with breast implants. The datasets were further refined by removing women without information regarding which side the cancer develops.

3. METHODOLOGY

3.1 Data Pre-processing

The images came in their processed format from the mammography machine, however, several further pre-processing techniques were applied. Otsu's segmentation was used to segment the background from the breast tissue in order to homogenise the background. Right mammograms were flipped horizontally to ensure they were all facing the same way. The images were min-max standardised to fit within the $[0, 1]$ range. For positive cases, only the side with cancer or that develops cancer were considered and no information from the contralaterals was used. For negative cases the breast side was randomly sampled.

Each image is split into N smaller, equally-sized patches $X = x_1, x_2, ..x_N$ where $x_i \in \mathbb{R}^{224\times224}$ and N is dependant on the breast size. We have elected to use 224×224 sized patches as this size is consistent with many

pretrained algorithms and ensures no images required rescaling. Patches with more than 50% background were deleted in order to remove those with too much background. Subtle, patch-level augmentations were applied to reduce the effect of overfitting: random rotations within 10 degrees, random translation within 5%, random scaling in the range of (0.9,1.1), and random shearing within 10 degrees of both the x and y axes. Patches were replicated across the three image channels.

3.2 Model

Figure 1 describes the full, end-to-end procedure of our approach with attention pooling. A pretrained ResNet-18[19] feature extractor is first applied to every instance in a bag associated to the images. The instances are then aggregated using one of two methods. The first is **average pooling** which computes the average across every feature vector in a bag producing an average pooled vector representation for the bag. It is described in Equation 1:

$$z = \frac{1}{N}\sum_{n=1}^{N} x_n \tag{1}$$

Whilst average pooling is a common method to combine instances it makes the assumption that every patch in a given bag holds equal importance. However, we know this is not true as areas that contain mammographic features pertinent to breast cancer such as masses or calcifications are more important than areas with only breast tissue. Recent literature[20] has promoted the use of attention-based pooling operators that can automatically assign importance to each patch based on learned information. The principle of attention based pooling is to find a weighted average of instances using fully-connected nets and pool based on the new averages. This would effectively solve the issue with variable patch importance as this is precisely what the network would be learning. The instance weights can also be used to enhance the transparency of the risk prediction model, by highlighting the patches that influence the decision the most.

Based on methods described by Ilse et al.[20,21] we have elected to use **gated-attention pooling**. This method is defined in Equations 2 and 3:

$$z = \sum_{n=1}^{N} a_n x_n \tag{2}$$

$$a_n = \frac{\exp\{w^T \tanh(V x_n^T) \odot \mathrm{sigm}(U x_n^T)\}}{\sum_{j=1}^{N} \exp\{w^T \tanh(V x_j^T) \odot \mathrm{sigm}(U x_n^T)\}} \tag{3}$$

where $\sum_{n=1}^{N} a_n = 1$ and w, V and U are parameters of the network. Softmax ensures the weights add up to 1 to resemble the weighted average. Finally, the pooled feature vector z is fed into a linear perceptron to make the final prediction. Both of these processes are described in Figure 2.

3.3 Experimental Details

As explained previously, the model is trained using the PROCAS-sdc dataset and the PROCAS-priors dataset is reserved purely for testing. Additionally, the model is tested using the contralateral dataset, which consists of the contralateral views of patients in the PROCAS-sdc test set. A summary can be seen in Figure 3.

The model is trained using 10-fold cross-validation with 15% of the training set reserved for validation. Both the CC and MLO views are fed into the model simultaneously, with both views from the same women in the same train/test split. During training, the negatives are undersampled 1 to 2 positive to negative which ensures a more even class balance. The model is trained using a learning rate of $l = 0.000007$ for 30 epochs. Each fold is restarted 3 times to avoid bad initialisations and the models are tested using the best performing epoch in the fold as measured by the validation set performance. The evaluation metric we are using is the AUC. Adam optimiser[22] is employed with a weight decay parameter of 0.00001 and negative log-likelihood as the loss function.

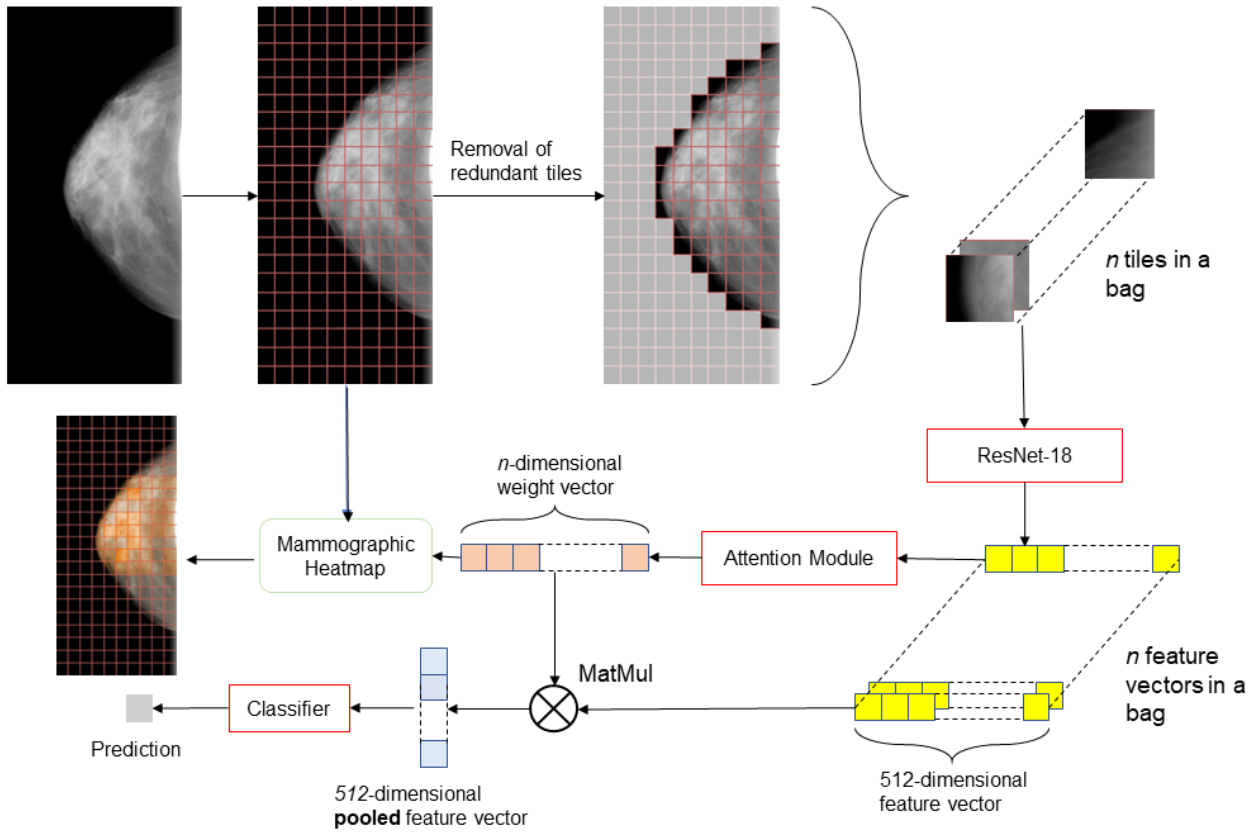


Figure 1. The end-to-end approach of our multiple instance learning model using attention.The process describes how the mammographic images is split into patches, fed into a CNN-based feature extractor and aggregated using attention. Output from the attention additionally enables the creation of saliency maps.

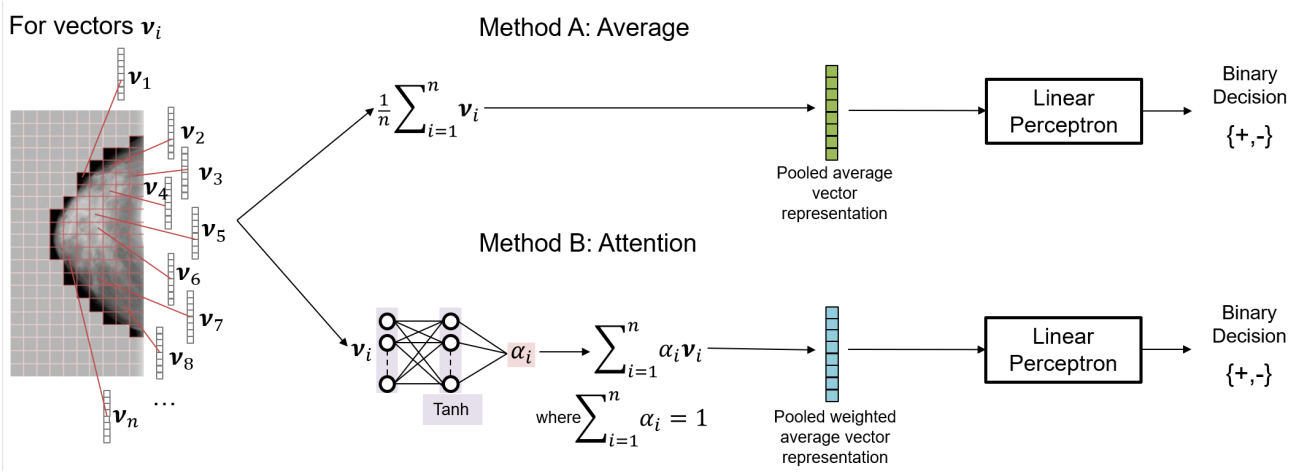


Figure 2. Explanation of the functionality of the two distinct aggregation methods. Method (A) combines the instances using average pooling. Method (B) combines the instances with attention based pooling using learned parameters.

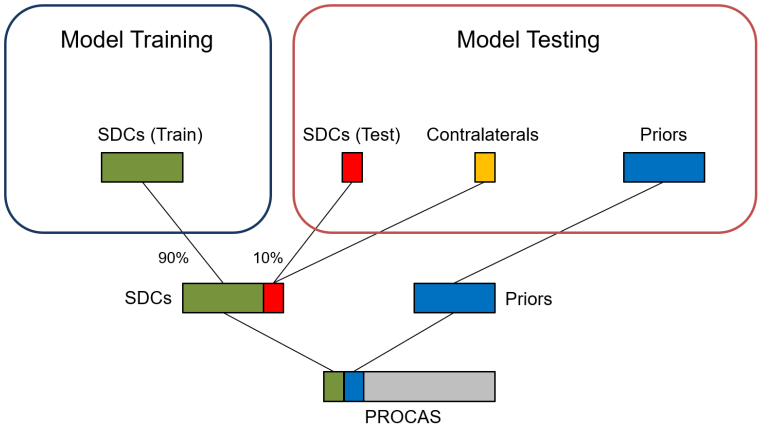


Figure 3. Breakdown of data use during model training and testing. PROCAS-sdc (SDC) is split into training and test sets. For the test sets, the contralaterals are also independently tested. PROCAS-priors is reserved exclusively for testing.

Table 1. AUC and 95% confidence intervals of average and attention based pooling methods for the three different datasets: PROCAS-sdc (SDC), PROCAS-sdc contralaterals (contra) and PROCAS-priors (prior).

	View	AUC(SDC)	AUC(contra)	AUC(prior)
Average	CC	0.534 (0.499,0.569)	**0.579** (0.544,0.613)	0.526 (0.489,0.564)
	MLO	0.502 (0.466,0.534)	0.545 (0.509,0.579)	0.496 (0.458,0.533)
	Combined	0.527 (0.492,0.561)	0.565 (0.530,0.600)	0.511 (0.473,0.548)
Attention	CC	0.769 (0.740,0.797)	0.534 (0.500,0.567)	0.627 (0.592,0.663)
	MLO	0.778 (0.748,0.808)	0.553 (0.518,0.586)	0.617 (0.583,0.651)
	Combined	**0.804** (0.777,0.830)	0.550 (0.516,0.584)	**0.635** (0.600,0.669)

4. RESULTS

Analysing the results Table 1, several patterns emerge. First of all, average pooling has failed to achieve statistically significant results for the SDCs with a combined AUC of 0.527 (0.492,0.561) and priors with a combined AUC of 0.511 (0.473,0.548), thus its no better than random guessing. Meanwhile, attention based pooling has substantially better results for these two categories with combined AUCs of 0.804 (0.777,0.830) and 0.635 (0.600,0.669) respectively. The CC and MLO views performance is similar, which is significant and reinforces the idea that we do not require to develop separate models for both views. Combining them generally improves the overall performance. Interestingly, average pooling has outperformed attention for the contralateral dataset only with an AUC of 0.579 (0.544, 0.613) achieving statistical significance.

Qualitative results are also available and can be seen in Figure 4 for the SDCs and Figure 5 for the priors. A radiologist was asked to point out the lesion location in given mammograms, and their annotations were then compared to our model's saliency maps. Unfortunately, we were unable to acquire supporting radiologist annotations for the priors as of yet, but we are working on resolving the issue. More zoomed-in diagrams are shown in Figures 6 and 7 which present the precise mammographic features the saliency maps delineate.

4.1 Discussion

There are several deductions we can make from our results. Firstly given how much more effective attention pooling is than average pooling suggests how instrumental it is to our approach. This may be attributed to the fact only a small comparative area of the breast in mammography is important to the classification, such as areas with visible masses, asymmetry or calcifications. Indeed looking at Figures 4 we can see our attention method giving significant weight to a small number of patches. It is clear that in these examples the attention module is picking out lesions when weighing the patch importance's. When compared against an independent radiologist assessment, our model was able to localise the lesions for these two patients.

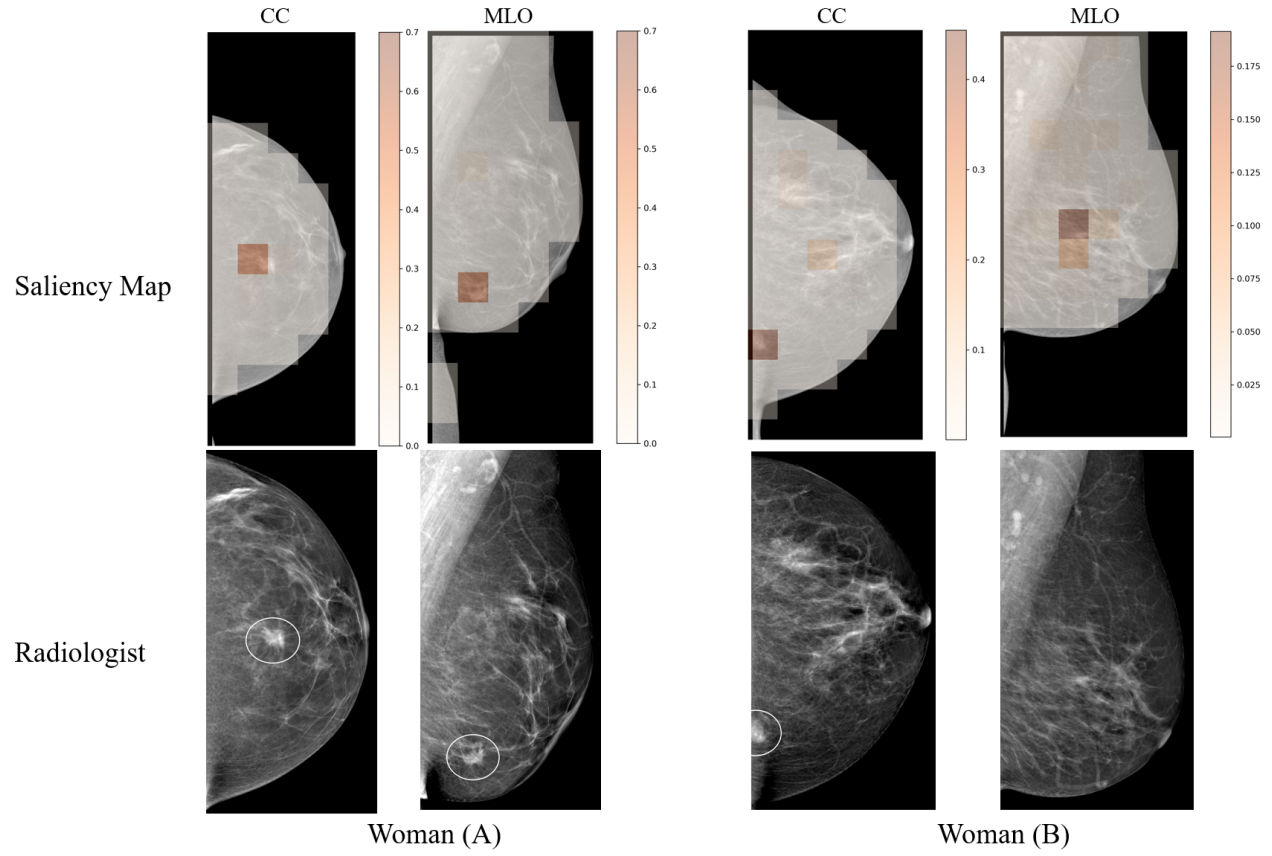


Figure 4. Saliency maps from our model against an independent radiologist annotations for two women. Woman (A) is cancer-positive with a visible lesion which our model was able to successfully identify. Woman (B) is also cancer-positive where the lesion is not visible in the MLO view as confirmed by a radiologist.

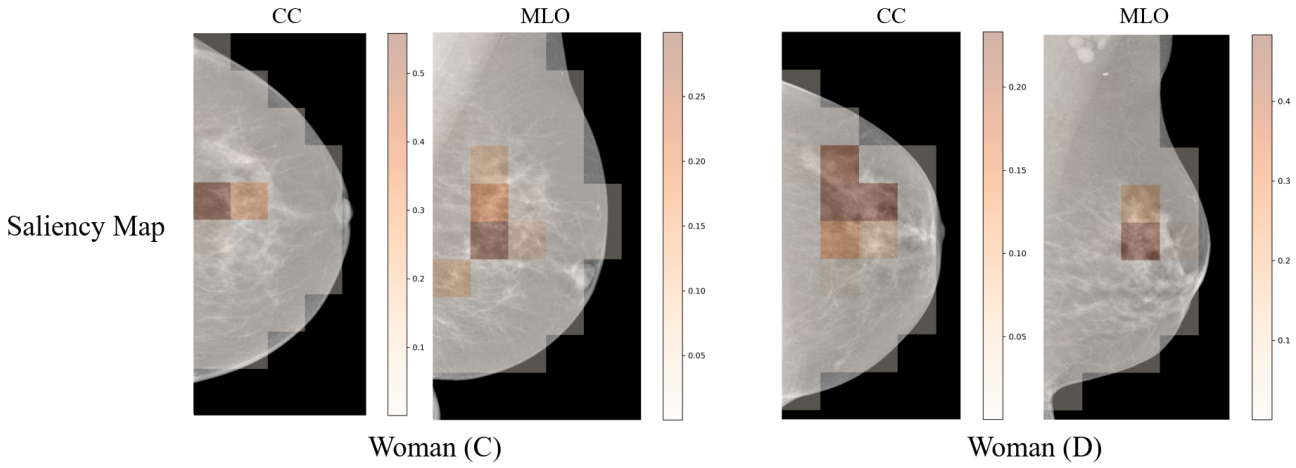


Figure 5. Saliency maps for the priors. Both women go on to develop breast cancer on a subsequent screen which our model was able to successfully predict.

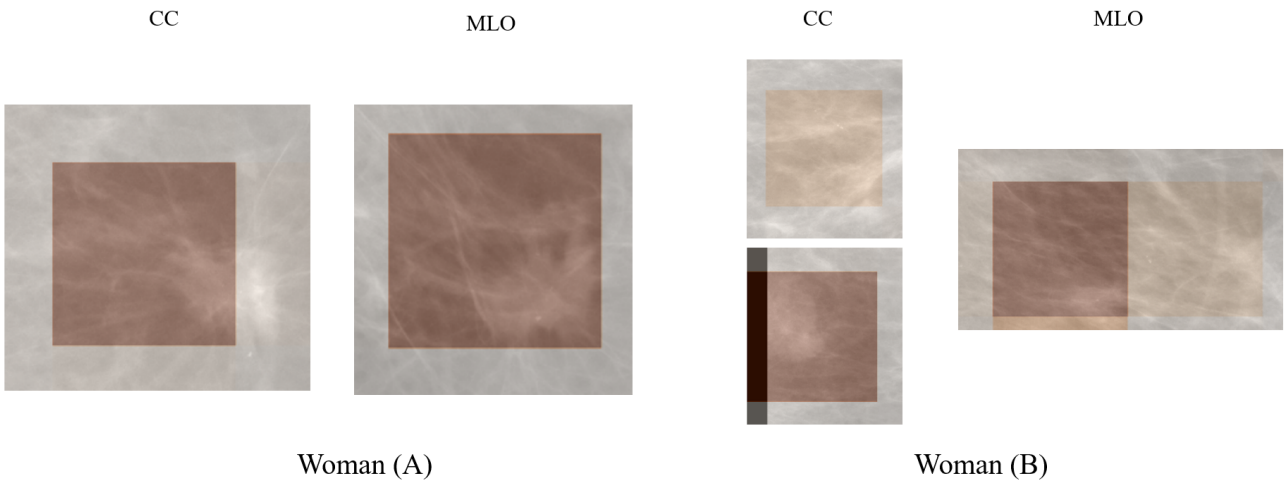


Figure 6. Zoomed-in patches from women A and B showing the detected masses corresponding to the radiologist annotations. For woman B, areas of high calcification density are also shown.

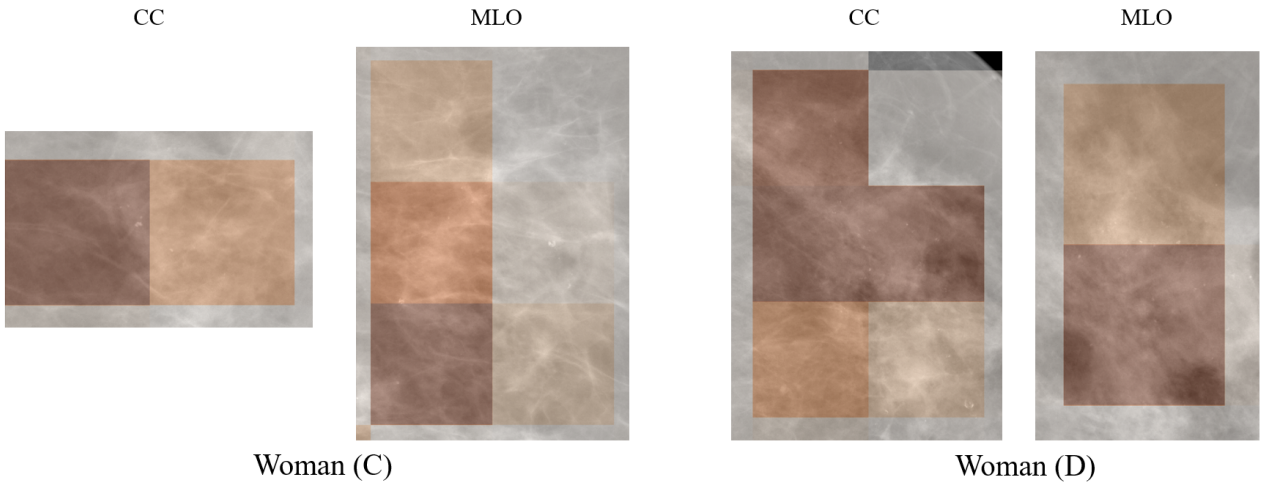


Figure 7. Zoomed-in patches from women C and D revealing the highlighted importance of calcification clusters.

What is notable about this, is that the model objective was to maximise how well the model could tell the positive and negative classes apart. The model independently learned to pick out the mammographic features without any prior annotations. We hypothesise this feature is a key reason as to why our model is able to extract signal from the priors despite being trained on PROCAS-sdcs. The model learned to recognise important mammographic features from the SDCs, similar to a radiologist, and successfully apply them to the priors.

Indeed studying Figure 5, the model is selecting regions with clusters of calcifications. It would be interesting to compare these regions against the locations where the future cancer develops, and we are working with radiologists to acquire these. Nevertheless, such saliency maps provide practical insight into the model and could potentially assist radiologists with future cancer diagnoses or missed lesions.

Another striking feature is that the attention module is able to localise the same mammographic features in the two views separately. In Figure 4 for woman B this is particularly notable as the lesion is not visible in the MLO view and our model was not able to pick it out, but both views have the same patch of calcifications highlighted with comparatively equal importance. Using both views in the same model has not hindered the performance significantly, however there is still a weakness in the way the views are combined. In Figure 4 for woman B the MLO view was ultimately classified as negative by our model since there is no visible signs of malignancy whereas a radiologist would make the decision based on both views.

The discrepancies between model performance for the PROCAS-sdcs and the priors are to be expected given that the priors are solving a more difficult problem. A screen-detected cancer by definition would have masses present which is the only marker the model needs to find, yet favourable performance on the priors suggests that the model also considers other factors, which is ultimately why transfer learning was successful. It is common for mammography based cancer detection models to look for candidate regions of interest that may contain a mass.[13] Whilst this is a sensible approach for detection specifically, it would not work on the priors due to a lack of visible masses and a wider range of notable features which is where our approach has a distinct advantage.

5. CONCLUSION

In conclusion, screen-detected cancers are a viable substitute when dealing with data scarcity for risk prediction. Models trained to detect breast cancer can be applied to breast cancer risk prediction models and necessitates further investigation. Multiple instance learning has successfully reduced the computational burden of the problem without any information loss. The proposed approach is independent of mammogram dimension or breast size, and does not rely on any annotations. The generated saliency maps could provide a useful glimpse into the "black box" of a deep learning model and potentially produce useful annotations mimicking that of a radiologist, but these claims necessitate additional systematic validation. Nevertheless, the proposed methods and their output justify the need to secure more annotated data.

5.1 Future Work

Access to annotated data will enable us to train more robust models and to systematically verify the accuracy and clinical utility of our saliency maps. For the priors, collecting and annotating future screen-detected cancers would allow us to retrospectively validate whether the saliency maps are correlated with future cancer mass locations.

The next avenue we wish to pursue is addressing the issue of separate views. Figure 4 demonstrated a potential weakness of our approach, and ultimately emphasized the need for two separate screen mammography views in the first place. Developing a model that can simultaneously analyse both mammographic views and produce a single informed decision is vital if we wish for our model to have clinical relevance alongside the screening programme. Fortunately, multiple instance learning provides an obvious research avenue by combining both mammograms into a unified bag of instances associated to a given woman. Such design would also produce more detailed saliency maps that automatically accentuate areas across views and sides.

We are also planning to improve on the current feature extractor. Aside from testing alternative architectures other than ResNet-18, self-supervised learning could be a compelling direction. Given the amount of unused data from PROCAS available, such data may find its purpose in self-supervised learning pretext tasks. Other options may employ the use of vision transformer.

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